

# Pulkit Verma

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## Research Interests

AI Safety, AI Planning, Analysis of Abstractions, and Action-Model Learning.

## Education

### Ph.D in Computer Science, Arizona State University

Advisor: Prof. Siddharth Srivastava [✉](#) | GPA: 4.0/4.0

Tempe, AZ

Exp. Fall 2023

### Master of Technology in Computer Science & Engineering, IIT Guwahati

Advisor: Prof. P. K. Das [✉](#) | Thesis: Resource Usage Analysis for Speech Recognition Techniques [✉](#) | GPA: 9.07/10

Guwahati, India

May 2015

### Bachelor of Engineering in Information Technology, SVITS, Indore

Advisor: Prof. Anand Rajawat | Capstone: Smart Analyzer - Predict and Cache Webpages based on Usage.

Indore, India

June 2011

## Publications

**Pulkit Verma**, and Siddharth Srivastava. "Learning Causal Models of Autonomous Agents using Interventions". In *IJCAI 2021 Workshop on Generalization in Planning*, 2021. (To appear)

**Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Learning User-Interpretable Descriptions of Black-Box AI System Capabilities". In *ICAPS Workshop on Knowledge Engineering for Planning & Scheduling*, 2021. [✉](#)

**Pulkit Verma**, Shashank Rao Marpally, and Siddharth Srivastava. "Asking the Right Questions: Learning Interpretable Action Models Through Query Answering". In *Proc. of the AAAI Conference on Artificial Intelligence*, 2021. [✉](#)

**Pulkit Verma**, and Pradip K. Das. "A Comparative Study of Resource Usage for Speaker Recognition Techniques". In *International Conference on Signal Processing and Communication*, 2016. [✉](#)

**Pulkit Verma**, and Pradip K. Das. "i-Vectors in Speech Processing Applications: A Survey". In *International Journal of Speech Technology*, 18(4), pp.529-546, 2015. [✉](#)

Mayank Gupta, **Pulkit Verma**, Tuhin Bhattacharya, and Pradip K. Das. "A Mobile Agents based Distributed Speech Recognition Engine for Controlling Multiple Robots". *2nd International Conference on Advances In Robotics*, 2015. [✉](#)

**Pulkit Verma**, Mayank Gupta, Tuhin Bhattacharya, and Pradip K. Das. "Improving Services using Mobile Agents-based IoT in a Smart City". In *2014 International Conference on Contemporary Computing and Informatics*, 2014. [✉](#)

## Professional Experience

### Arizona State University

Tempe, AZ

#### Graduate Research Associate

Aug. 2018 - (present)

- Investigating the minimal set of interfaces in an AI system that would enable a user to assess and understand the limits of its safe operability.
- Developed a framework which generates interrogation policies to derive an interpretable model of an AI agent.
- Co-designed the experiments showcasing possible use-cases of an NSF project on training lay people to use adaptive AI systems which are safe, robust and reliable.

### NetApp Inc.

Bengaluru, India

#### Storage Efficiency Developer

Jul. 2015 - Jul. 2018

- Workload Prediction*: Predicted workload on a storage volume based on I/O patterns for All Flash systems. This helped in optimizing data storage and management while keeping the nature of data private to the users.
- File and Volume Clones*: Developed features and managed issues related to file and volume clones on UNIX-based NetApp proprietary WAFL file system to improve the storage efficiency of enterprise data servers.
- Co-developed a file system component which prevented data loss if files and volumes are cloned or modified when cloud backup is offline.

### Tata Consultancy Services Ltd.

Pune, India

#### iOS Application Developer

Mar. 2012 - Aug. 2013

- Built interactive applications for iPhone and iPad and developed tools to analyze the user investment portfolio.
- Performed cross-platform application development of mobile applications supported on both Android and iOS.

## Teaching Experience

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### Guest Lecturer

Arizona State University

Tempe, AZ

2020-2021

- CSE 574 Planning and learning Methods in Artificial Intelligence, 2 Lectures, 39 students, Spring 2021.
- CSE 471 Introduction to Artificial Intelligence, 1 Lecture, 118 students, Fall 2020.

### CSE 571 Artificial Intelligence [Graduate-level Course]

Arizona State University

Tempe, AZ

Fall 2019

- Delivered tutorials on Constraint Satisfaction Problems, PDDL, First Order Logic, and Dynamic Bayesian Networks.
- Co-created the software testbed used to support the course projects. Also created and graded homework assignments and exams.
- Mentored project groups working on class projects.

### CS 566 Speech Processing [Graduate-level Course]

Indian Institute of Technology Guwahati

Guwahati, India

Fall 2014

- Co-developed a custom C++ API to use a voice recording tool that students used in course projects.
- Mentored students working on class projects and homeworks.
- Graded homework assignments and exams.

### Undergraduate Lab. Courses

Indian Institute of Technology Guwahati

Guwahati, India

2013-2015

- TA for the lab courses: CS 513 Programming Lab.; CS 462 Graphics Lab.; and CS 243 Software Engineering Lab.
- Graded homework assignments and exams.

## Academic Projects

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### Analysis of Gated Updates in Value Iteration Networks

CSE 574 | Planning and Learning Methods in AI | ASU

- Replaced non gated RNN update in Value Iteration Networks with gated updates; long short term memory (LSTM), gated recurrent units (GRUs), and peephole-LSTM.
- Demonstrated that peephole-LSTM performs best for time-bound scenarios, whereas GRUs are recommended for non-time-bound scenarios.

### Automatic Image Captioning using Topic Modeling

CSE 569 | Fundamentals of Statistical Learning | ASU

- Generated image caption topics using correspondence Latent Dirichlet Allocation with SIFT image features.
- Used Gibbs sampling to calculate posteriors instead of variational inference.
- Tested on Corel-5K, ESP game, and MIR flickr datasets.
- Showed that the accuracy increases as we increase number of topics till a value, and then starts increasing again.

### Artificial Immune System Based Obstacle Avoidance Robot

CS 666 | Mobile Robotics | IIT Guwahati

- Designed a robot which learns to avoid the “dynamic obstacles” using “Clonal Selection” (Immune System concept similar to Reinforcement Learning).
- Experimented with 3 scenarios; 2 robots following fixed paths, 1 robot with one periodically moving obstacle, and 2 robots with unconstrained motion within a region.
- Showed that reward shaping is necessary for robots to learn to avoid obstacles in constrained environments.

### Robot Cooperation through Stigmergic Communication

CS 666 | Mobile Robotics | IIT Guwahati

- Modeled the robots to perform sequential tasks by automatic-synchronization using stigmergic communication.
- Used the “Typhon” mobile agent framework to create a two level decentralized “ring” network.
- Sequentially increased the homogeneous cooperating robots in the network till task was completed autonomously.
- Demonstrated the limitations of scalability of “Typhon” which were then addressed in its next version “Tartarus”.

## Resource Usage Analysis for Speech Recognition Techniques

Masters Thesis | IIT Guwahati

- Analyzed the impact of i-vectors (a projection of speech feature into total variability space) on devices having low resources like memory, storage space and power for real time speech recognition.
- Showed that i-vectors are better as compared to GMM-UBM and Joint Factor Analysis based approaches for low resource devices.

## Predicting Meal Times for Automated Insulin Delivery

CS 565 | Mobile Computing | ASU

- Correlated the noisy real data sampled at different frequencies from continuous glucose monitors (CGMs) and insulin bolus.
- Experimented with SARIMA and RNN for training the time series models for glucose level prediction. Then used the predicted glucose levels and insulin bolus data to infer meal intake times.
- Showed that SARIMA predicts glucose levels with better accuracy than RNN in lesser time.

## Smart Analyzer: Predict and Cache Webpages based on Usage

B.E. Capstone Project | SVITS, Indore

- Designed a system that analyzed the patterns of access of websites through a Squid proxy server, and predicted the webpages to be cached.
- Used Apriori algorithm to find correlation between the webpages accessed over different time periods and predict the webpages to be cached depending on the current webpage requested.

## Service

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- 2021 **Program Committee Member**, ICAPS 2021 Workshop on Explainable AI Planning (XAIP'21) [↗](#)
- 2021 **Program Committee Member**, IJCAI 2021 Workshop on Generalization in Planning (GenPlan'21) [↗](#)
- 2021 **Volunteer**, 9th International Conference on Learning Representations (ICLR 2021) [↗](#)
- 2021 **Research Grants Reviewer**, Graduate and Professional Student Association, ASU [↗](#)
- 2021 **GPSA Awards Reviewer**, Graduate and Professional Student Association, ASU [↗](#)
- 2019 **Volunteer**, Southwest Robotics Symposium 2019, Tempe, AZ [↗](#)
- 2018 **Volunteer**, 16th Int. Conf. on Principles of Knowledge Representation & Reasoning (KR 2018) [↗](#)
- 2014 **Volunteer**, National Workshop on GPU Programming and Applications, IIT Guwahati [↗](#)

## Honors & Awards

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- Mar 2021 **Graduate College Travel Awards**, ASU (for AAMAS 2021, ICLR 2021, ICML 2021, IJCAI 2021)
- Jan 2021 **GPSA Travel Grant**, Arizona State University (for AAAI 2021)
- Oct 2018 **Fulton Schools Experiential Learning Grant**, Arizona State University
- Aug 2013 - May 2015 **Post Graduation Fellowship**, All India Council for Technical Education, Government of India

## Other Affiliations

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- Nov 2020 - (present) **Affiliated Graduate Student**, Center for Human-Compatible Artificial Intelligence, UCB [↗](#)
- Aug 2018 - (present) **Graduate Student**, Autonomous Agents and Intelligent Robots (AAIR) Lab, ASU [↗](#)
- Jul 2016 - Dec 2016 **Online Contributor**, Stanford Scholar Initiative [↗](#)
- Jan 2014 - May 2015 **Graduate Student**, Robotics Lab, IIT Guwahati [↗](#)

## References

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- Prof. Siddharth Srivastava**  
siddharths@asu.edu Assistant Professor, School of Computing, Informatics, and Decision Systems Engineering, Arizona State University, USA
- Prof. Pradip K. Das**  
pkdas@iitg.ac.in Professor, Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, India
- Prof. Shivashankar B. Nair**  
sbnair@iitg.ac.in Professor, Department of Computer Science and Engineering, Indian Institute of Technology Guwahati, India
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